



Module: Electronics, system components

Corrected Series of TD n° 03

Exercise 1:

- | | |
|----------------|----------------------|
| 1) Screen | 5) Expansion card |
| 2) Motherboard | 6) Central unit case |
| 3) Processor | 7) Keyboard |
| 4) RAM | 8) Mouse |

Exercise 2:

1) The motherboard is the foundation that makes the components and technologies of our computers work. When assembling a computer, we first fix the motherboard in the central unit case then connect all the electronic elements to it: processor, memory, graphics card, PCI expansion card, but also the hard drives, CD/DVD player, keyboard, mouse, speaker...

2) The main components of the motherboard:

- The chipset
- The clock
- CMOS
- The battery or accumulator battery
- The system bus or Front Side Bus (FSB)
- The memory buses
- The expansion bus (input/output bus)
- The input/output panel
- Motherboard connectors

3) Some characteristics of the baseboard are listed below:

- Motherboards relative have a different connection and compatibility which again differs from chipset to chipset and company to company.
- Each type of mainboard design, case, size, power supply is made according to the requirement. So, they all are compatible to work properly.
- Each type of mainboard design supports working only with specific types of processors and a few types of memory.

These are the Features of a Good Motherboard.

4) The size factor of a motherboard means the geometry, dimensions, layout and electrical characteristics of the motherboard.

5) The chipset establishes and manages dialogues between the different components of the computer (processor, memory, peripherals, etc.).

6) The chip that manages the dialogues between the processor and the RAM is the North Bridge.

Exercise 3:

1) It is possible to start a computer without a motherboard

B- False

2) The element that connects all the elements constituting a computer is:

C. The motherboard

3) The role of the processor in a computer is: (choose the correct answers)

B. To carry out the instructions

C. To order all units

E. To control the progress of all operations

4) The command-and-control unit

D. Produces orders

5) Instruction register stores the address of the instruction being executed

B- False

6) The state register:

A. Allows you to store indicators on the state of the system (withholding, overflow, etc.).

7) The accumulator

B. Stores the address of the instruction currently being executed

8) Indicate the correct proposition among the following:

D. The UAL performs logical operations and arithmetic operations.

9) The arithmetic and logic unit are essentially composed of:

C. Arithmetic and Logic Unit, State Register, Accumulators

10) MIPS is a unit of measurement:

C. Power: Millions of Instructions Per Second

Exercise 4:

1) Convert the following capacities to byte: 13 MB, 2.3 GB, 1.435 TB, 123 KB

2) Convert the following capacities to MB: 13 GB, 9 TB, 1400 KB, 23 KB, 1,435,897 B

Conversion table

1 byte = 8 bits (not to be confused with byte (English) = octet (French).)

1 kilooctet (kB) = 1 024 Byte = 2^{10} Byte

1 megaoctet (MB) = 1 024 kB = 1 048 576 Byte = 2^{20} B

1 gigaoctet (GB) = 1 024 MB = 1 073 741 824 Byte = 2^{30} B

1 teraoctet (TB) = 1 024 GB = 1 099 511 627 776 Byte = 2^{40} B

1 petaoctet (PB) = 1 024 TB = 1 125 899 906 842 624 Byte = 2^{50} B

1) Conversion to byte

13 Mo = $13 \times (1024)^2 = 13631488$ octet

2.3 Go = $2.3 \times (1024)^3 = 2469606195$ octet

1.435 To = $1.435 \times (1024)^4 = 1577799185858.56$ octet

123 KB = $123 \times 1024 = 125952$ octet

2) Conversion to MB

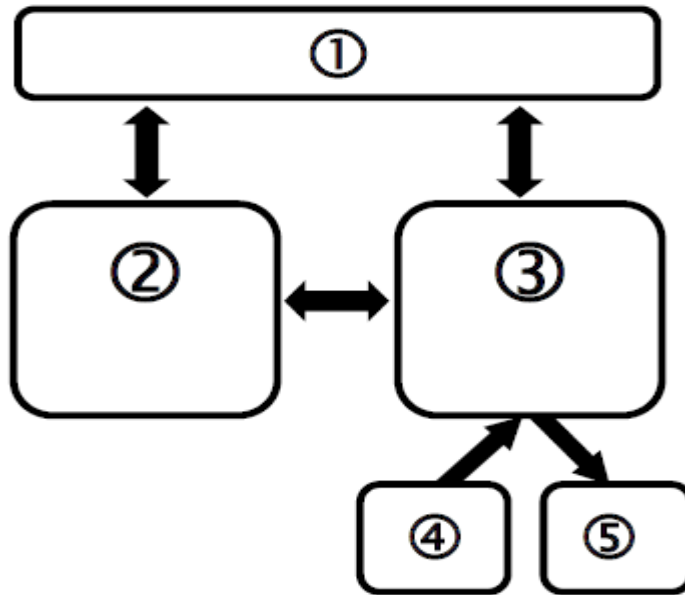
13 Go = $13 \times 1024 = 13312$ Mo

9 To = $9 \times (1024)^2 = 9437184$ Mo

1400 Ko = 1,36 Mo

23 Ko = 0.022 Mo

Exercise 5:



1) The parts of Von Neumann's architecture:

1. Memory
2. Control unit
3. Arithmetic and logic unit
4. Input
5. Output

2) A processor is an electronic component in which all the components have been sufficiently miniaturized to be grouped in a single box (it is a very large-scale integrated circuit). Often made of silicon, which brings together several million interconnected elementary transistors. Its role is to interpret and execute program instructions.

3) The two main processor manufacturers are Intel and AMD.

Model for Intel: Celeron, Pentium, Core

Model for AMD: Sempron, Athlon, Turion, Phenom

4) A processor has a heat sink on top to dissipate heat to prevent its circuits from melting.

5) The different components of the processor are:

- **The arithmetic and logic unit (ALU)** where arithmetic and logic operations are performed.
- **The command or control unit** which directs the operation of all other units (UAL, memory, inputs/outputs) by providing them with control signals; its circuits generate the signals necessary to execute each instruction in a program.
- **Registers** responsible for temporarily storing the various information to be processed.
- **Internal buses** allowing the exchange of information between the three previous elements.

6) **By size:** Registers < L1 cache < L2 cache < RAM < CD < DD.

By speed: Registers > L1 cache > L2 cache > RAM > DD > CD.

The smaller memory size, the faster it is.

7) Random Access Memory, called RAM, is the computer memory in which a computer places the data during their processing, in which the operating system, the fonts of characters, etc.

Its two main characteristics are:

- **Speed of access (Access time):** essential to quickly deliver data to the processor.
- **Volatility:** it characterizes the permanence of information in a memory. A volatile memory loses its contents when the power is turned off.

8) Static random-access memory (SRAM)

Dynamic random-access memory (DRAM)

DRAM (Dynamic RAM) is based on capacitors that must be refreshed, while **SRAM (Static RAM)** is based on transistors that do not require refreshing. This refresh time means that DRAM is slower than SRAM but is less expensive because it uses fewer transistors. They are both volatile memories.

The more expensive but faster SRAM is used in cache memories while DRAM is used in main memory. The DRAM only retains the written information for a few milliseconds: the memory controller is therefore obliged to regularly reread each cell and then rewrite the stored information in order to guarantee its reliability; this recurring operation is called “refreshing”.

Despite these refresh constraints, this type of memory is widely used because it is much cheaper than static memory. Indeed, the elementary memory cell of the DRAM is very simple and requires little silicon.

9) We distinguish the following forms:

- **ROM:** its data is introduced directly during its manufacture. It is impossible to modify it.
- **PROM:** Programmable blank ROM. However, it can only be programmed once.
- **EPROM:** programmable and erasable ROM. It can be erased by ultraviolet and programmable at will.
- **EEPROM:** programmable and electrically erasable ROM.
- **FLASH memory:** derived from the EEPROM but with a smaller capacity. However, this capacity is evolving day by day. It is found in particular as storage memory in digital cameras (Compact Flash, Smart Media, etc.).

Good luck
Dr. Nadjat Azzaoui